Weekly Update

The TSB-AD-U leaders and the 12 views

Wavelet_AD weekly

28 October 2026

TSB-AD-U

48

development series (Tuning)

350

held-out series (Eva), run once

score every time point $\rightarrow$ VUS-PR

Protocol: data, metric, comparisons

Data

- series $x_1, \ldots, x_n$; training stretch $x_{1:T}$, treated as normal (T from the file name)
- x z-normalised with the training stretch's mean and std
- a score for *every* point, training stretch included (as the official runner)
- Tuning: 48 series, all design choices; Eva: 350, run once on a frozen method (2 looks used: 7 and 22 Oct)

Metric

- VUS-PR: precision–recall area, averaged over tolerance buffers around each labelled range
- buffer from `find_length_rank(rank=1)` on the whole raw series (TSB-AD code)
- our numbers: TSB-AD's `generate_curve` via Hankel_AD `calculate_metrics`

Comparison

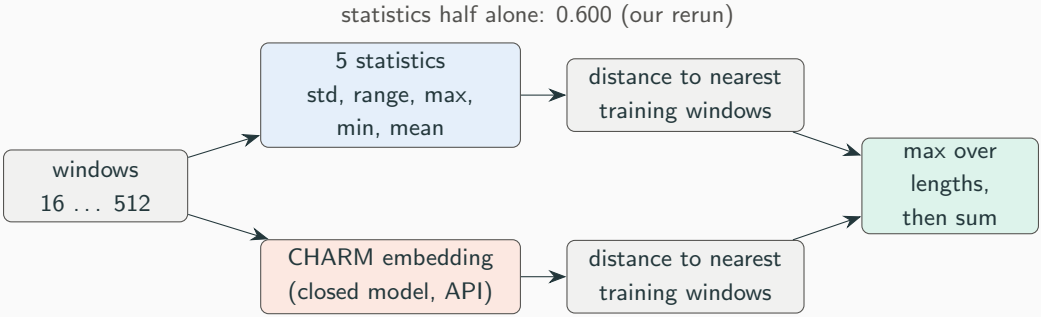
- paired over series: mean of per-series differences
- 95% bootstrap interval (10 000 resamples, seed 0); better/worse counts

Anomaly length: two ways to bin the series

- a = median labelled anomaly length (samples) L = period from the training stretch, `find_length_rank` (autocorrelation peak; 125 if none or > 300: 7 of 48 Tuning series)

a **in samples** (used here)	1–3	4–32	33–256	> 256
Tuning / Eva series	11 / 57	10 / 57	18 / 140	9 / 96
a/L (DG_AD's axis)	< 0.03, 0.03–0.1, 0.1–0.5, 0.5–2, > 2			
Tuning / Eva series	7/29, 4/25, 7/48, 18/120, 12/128			

Leader 1: CHARM_kNN 0.678 (authors')



CHARM_kNN: the method in detail

Statistics half (no model). For each window length $w \in \{16, 32, 64, 128, 256, 512\}$, every start s (stride 1):

$$f_w(s) = \big[\,\mathrm{std},\ \max - \min,\ \max,\ \min,\ \mathrm{mean}\,\big](x_{s:s+w-1}) \in \mathbb{R}^5$$

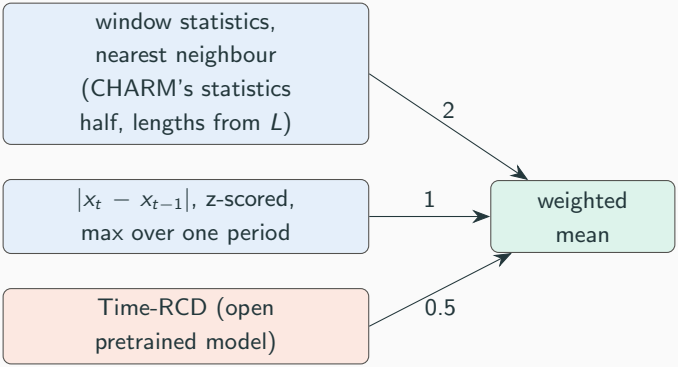
1. standardise each of the 5 features by the training windows' mean and std (reference capped at 10 000, random)
2. window score = distance to its $k = 3$ nearest training windows in $\mathbb{R}^5$
3. point score $A_w(t)$ = mean over the windows covering t
4. $S_{\text{stat}}(t) = \max_w z(A_w)(t)$, z = z-score over the whole series

Embedding half. Same recipe on CHARM embeddings (last layer, pooled over time) at $w \in \{64, 128, 256\}$.

Final: $S = S_{\text{stat}} + S_{\text{emb}}$, then min–max scaled.

Statistics half alone: 0.600 (DG_AD rerun of `Run_CHARM.py`, Eva). Training windows are their own nearest neighbour (self-match): +0.012 on that half. Scales, combiner and readout adopted on Eva gains.

Leader 2: Syntropic 0.669 (authors')



Syntropic: the method in detail

p = `find_length_rank` period. Each component is robust-z-normalised over the whole series, then

$$S(t) = \tfrac{1}{3.5}\big(2\,\mathrm{rz}(\mathrm{WSK}) + 1\,\mathrm{rz}(\mathrm{DiffZ}) + 0.5\,\mathrm{rz}(\mathrm{RCD})\big)(t)$$

1. **WSK**: CHARM's 5 statistics at $w \in \{\max(8, \lfloor p/2 \rfloor),\ p,\ 2p,\ 16,\ 64,\ 256\}$, dropping any shorter than 4 or longer than a quarter of the training stretch, $T/4$ (but always allowing up to 8); standardised by training windows (cap 20 000, evenly spaced); window score = mean distance to the 3 nearest training windows; point = **max** over covering windows; $\max_w \mathrm{rz}(\cdot)$
2. **DiffZ**: $d_t = |x_t - x_{t-1}|$, $z_t = \dfrac{d_t - \mathrm{med}(d_{1:T})}{1.4826\,\mathrm{MAD}(d_{1:T})}$, then a running max over p samples
3. **Time-RCD**: open pretrained Transformer (patch 16, 8 layers), zero-shot; per-step probability of the "anomaly" class; windows of 15 000; trained on synthetic series (per its paper)

Reproduced by DG_AD's review: 350/350 Eva series within 2.2×10^{-6}. Without Time-RCD: 0.610. Tuning: 0.653 (our run). Weights fixed after one Eva score was seen.

What the leaders share

nearest neighbour

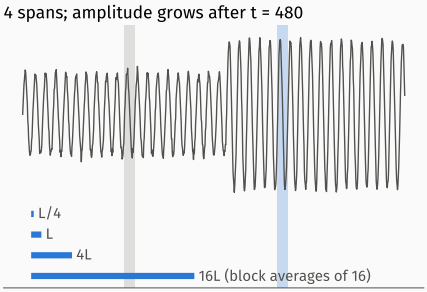
$\times$

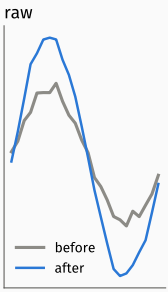
many window lengths

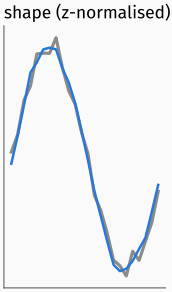
$+$ one pretrained model

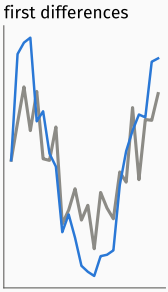
The 12 views

3 views × 4 spans

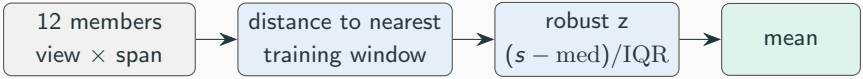








One score from 12



no training · no weights · only setting: the period L

The 12 views: the method in detail

For member (view v, block size r, window length ℓ), span $S = r\ell$:

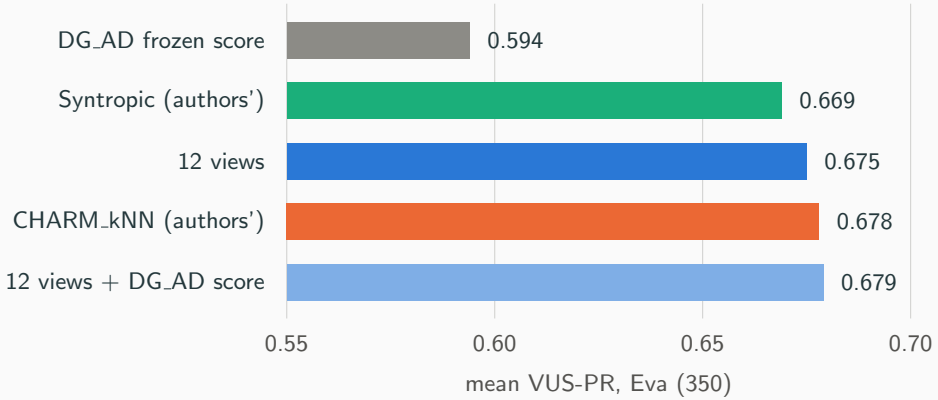
1. input: x (raw, shape) or $\Delta x_t = x_t - x_{t-1}$ (diff)
2. block means $m_r(t) = \frac{1}{r}\sum_{i<r} x_{t+i}$; window $W_t = \big(m_r(t), m_r(t{+}r), \ldots, m_r(t{+}(\ell{-}1)r)\big)$, stride 1
3. shape only: $W_t \leftarrow (W_t - \mu(W_t))/\max(\sigma(W_t), 10^{-6})$; μ, σ: mean and std of the window's own ℓ values
4. $s(t) = \min_{u \in \text{train}} \|W_t - W_u\|_2$; Theiler window: for training t, skip $|u - t| < S$ (overlapping windows)
5. point score: Gaussian-weighted mean of covering windows, width $S/4$
6. $\mathrm{rz}(s) = (s - \mathrm{med}\, s)/\mathrm{IQR}(s)$

Final score: mean of the 12 robust z-scores.

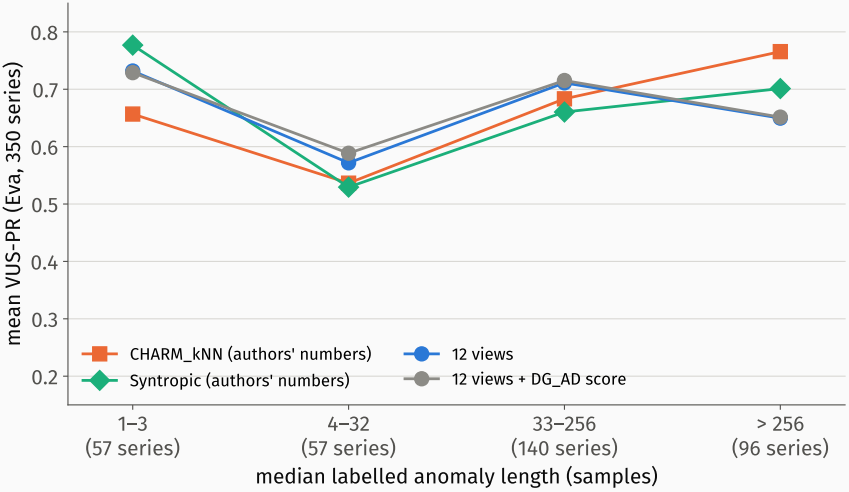
r	ℓ	span
1	$\max(2, \lfloor L/4 \rfloor)$	$L/4$
1	L	L
4	L	$4L$
16	L	$16L$

$\times$ 3 views = 12 members. A member with fewer than 10 training windows is dropped. No training, no weights. Code: `wavelet_ad/views.py`.

Level with the leaders



Where the leaders still win: long anomalies



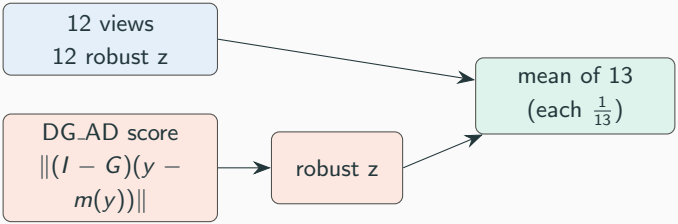
DG_AD's frozen score, in detail

- windows $y \in \mathbb{R}^L$ of length L (divided by $\sqrt{L}$); the training windows form the "healthy set"
- two networks, trained only on noisy copies of training windows (noise level $\varepsilon = 0.64/L$):
  - **denoiser** $m(y)$: maps a window back towards the healthy set
  - **metric network**: its output $G(m)$ projects onto the healthy set's local directions
- score of a window:

$$\left\| (I - G(m))\,(y - m(y)) \right\|$$

  the part of the denoising residual pointing *across* the healthy set
- point score: Gaussian readout over the window; frozen on 7 Oct; Tuning 0.532, Eva 0.594

Adding DG_AD's learned score: a 13th member



$$S(t) = \frac{1}{13}\Big(\sum_{v=1}^{12} \mathrm{rz}(s_v)(t) + \mathrm{rz}(s_{\mathrm{DG_AD}})(t)\Big)$$

	Tuning (48)	Eva (350)
13th member − 12 views	**+0.011** [+0.003, +0.019], 23/13	+0.004 [−0.000, +0.009], 175/125
in place of raw-L member − 12 views	+0.006 [−0.003, +0.015], 23/17	+0.001 [−0.004, +0.006], 169/130